

Dugongs and manatees

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Sirenia
FAMILIES	2
SPECIES	4

Sirenians—the dugong and manatees—are large, slow-moving creatures with a streamlined body. They are the only marine mammals that feed primarily on plants. Sirenians must rise to the surface to breathe, but they can remain submerged for up to 20 minutes. Even though they have no enemies apart from

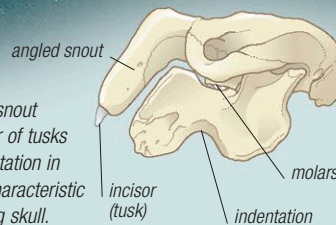
humans, sirenians may number fewer than 150,000—making them among the least abundant of any mammal order.

Anatomy

Sirenians have paddlelike front limbs and a flat tail to aid propulsion. Their skin is thick and tough, and they have a relatively small brain. Due to the large volume of gas given off during the digestion of plant matter, sirenians are highly buoyant. To compensate, their bones are heavy and dense. Underwater, sirenians can close their nostrils. They lack well-defined eyelids and protect the eye surface with secretions and by drawing the nictitating membrane across the eye.

DUGONG SKULL

The steeply angled snout bearing a single pair of tusks and a marked indentation in the lower jaw are characteristic features of a dugong skull.



MOTHER AND CALF

Sirenians are slow breeders: usually only one calf is produced every 2 years. The “mouthing” contact between this manatee mother and calf helps preserve the family bond.

Feeding

Sirenians strip vegetation using their large, mobile upper lip. Food is then crushed between horny plates on the front part of the palate and on the lower jaw, and finally ground between the teeth.



FORAGING ON THE SEA BED

Sirenians, such as this dugong, frequently search the sea bed for the rhizomes (underground stems) of sea grasses, which have high concentrations of carbohydrates.

CONSERVATION

In the past, sirenians were hunted extensively for their meat, hide, and oil. Today, they are vulnerable to injury or death caused by boat propellers (below), fishing nets, and the pollution of coastal waters.



Dugong dugon

Dugong



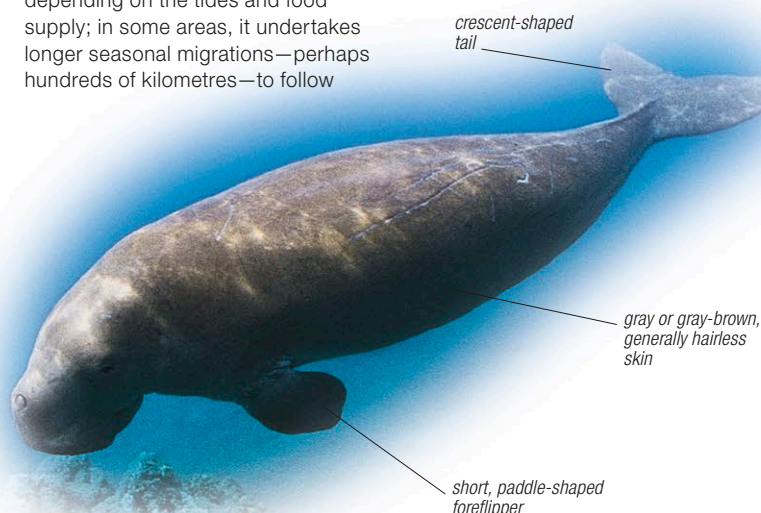
Length 6½–11 ft (2–3.3 m)
Weight 1,257 lb (570 kg)
Social unit Group
Status Vulnerable

Location E. Africa, W. Asia, S. Asia, S.E. Asia, Australia, Pacific islands



The dugong has a crescent-shaped tail, and short foreflippers. Primarily diurnal, it moves regularly each day between on- and offshore areas depending on the tides and food supply; in some areas, it undertakes longer seasonal migrations—perhaps hundreds of kilometres—to follow

marine plant growth and avoid cold-water currents. Dugongs form transient herds averaging 10–20, exceptionally 100 or more, with little social structure. Members may gather to intimidate and butt predators such as sharks. Males compete for females by sounds and pushes. Courtship and mating (which is monogamous) are similarly auditory and tactile. One offspring, up to 4 ft (1.2 m) long and 77 lb (35 kg) in weight, is born after a gestation of 14 months. It is cared for by the mother, with help from older siblings and female relatives, and is weaned by 18 months. A dugong may live for about 60 years.



crescent-shaped tail

gray or gray-brown, generally hairless skin

short, paddle-shaped foreflipper

Trichechus manatus

West Indian manatee



Location S.E. USA to N.E. South America, Caribbean



Best known of the 3 manatee species (the other 2 being the Amazonian and West African), the West Indian manatee lives along shallow shores and estuaries, and in nearby rivers and freshwater lagoons. Groups of up to 20 swell to 100 or more where food is plentiful; however, there is little cohesion, and individuals come and go, ranging widely. Reproduction is similar to the dugong's (see left), although the manatee is polygynous.

Length 8¼–9¾ ft (2.5–3 m)
Weight Up to 1½ ton (1.6 tonne)
Social unit Variable
Status Vulnerable

FOOD INTAKE



Feeding occurs from the surface down about 13 ft (4 m). The manatee holds food with its flippers, and directs it into the mouth using its flexible lips. Daily food intake is up to a quarter of its body weight, and may include a few fish (for protein).

BLUNT FRONT

Like other sirenians, this species has tiny eyes and no external ear flaps. Its gray or gray-brown skin, paler below, may harbor growths of algae.

